

PERTEMUAN 11

SEVEN LAYERS OSI

Deskripsi Model OSI

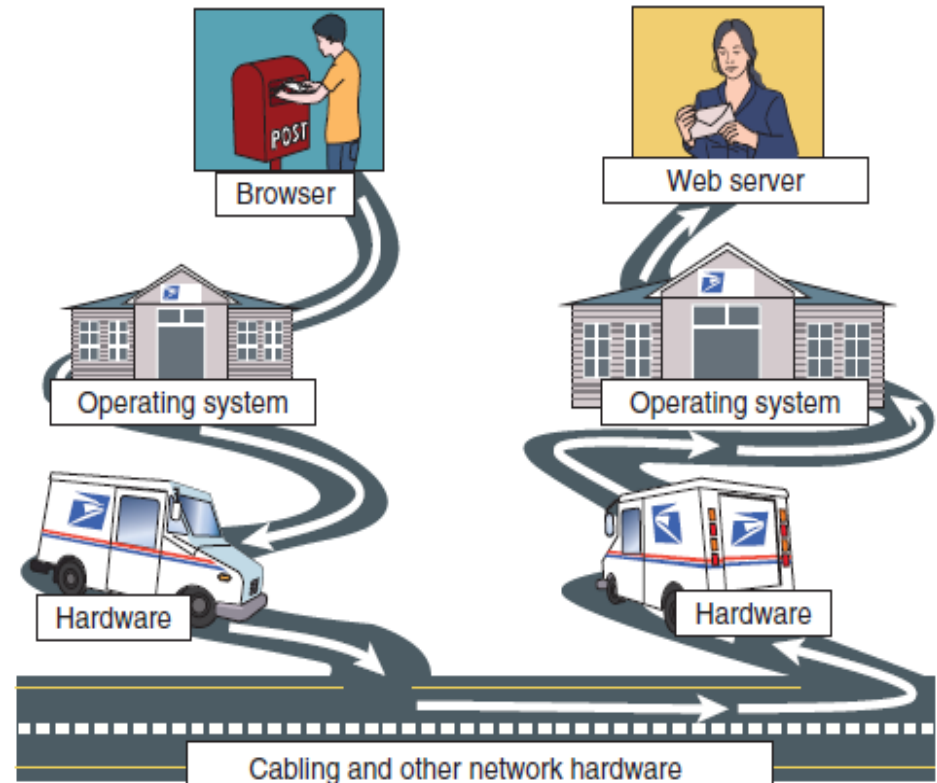
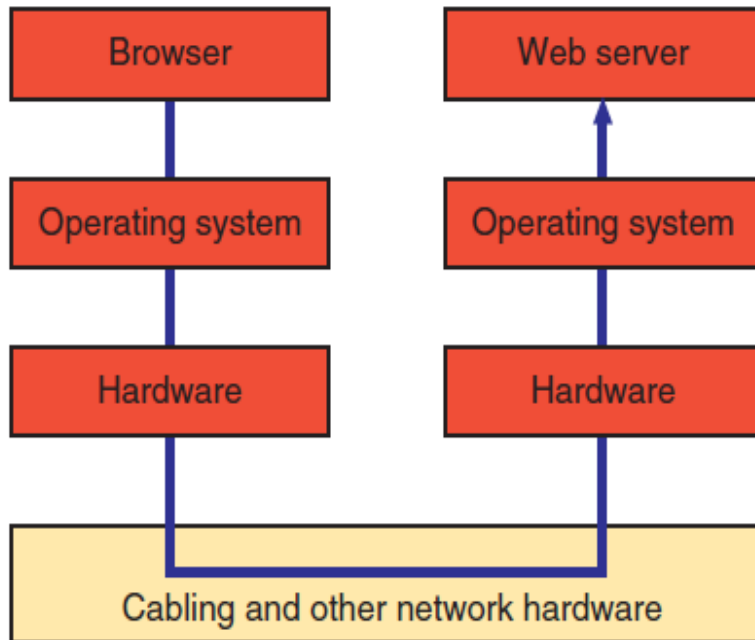
- Merupakan suatu model yang mengilustrasikan proses dan teknologi yang terkait dalam proses pengiriman data.
- Dipergunakan sebagai media komunikasi antar stakeholder atau praktisi jaringan komputer (network admin, teknisi hardware, programmer, network engineer) dalam menggambarkan fungsi-fungsi teknologi jaringan
- Berperan penting dalam pemecahan masalah dalam jaringan komputer, dengan pendekatan per lapisan (layered-approach)

Tujuan Model OSI

- Tujuan OSI ini adalah untuk membuat standar aturan komunikasi sehingga dapat terjalin interkomunikasi dari sistem yang berbeda tanpa memerlukan perubahan yang signifikan pada hardware dan software

Deskripsi Model OSI (2)

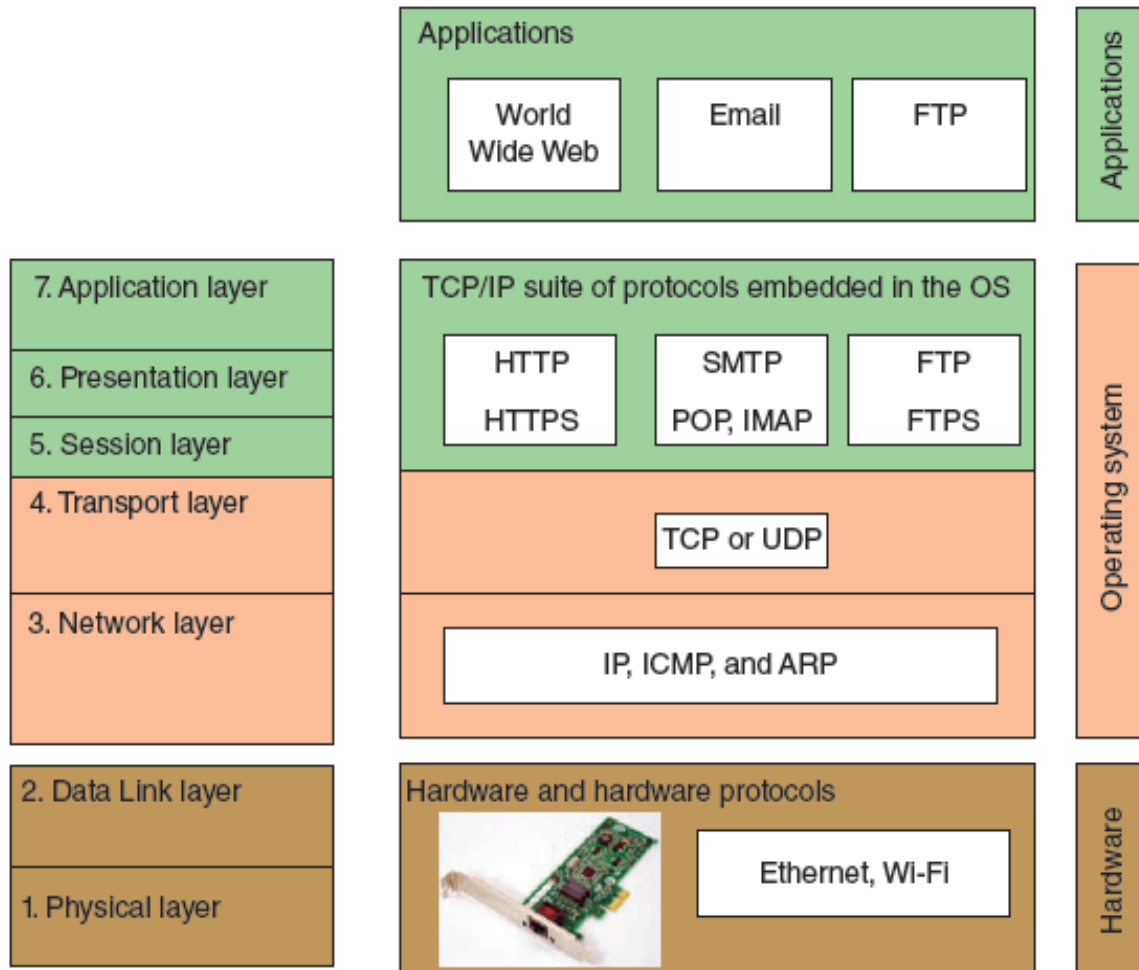
Analogi Pengiriman Data & Pengiriman Surat



Model OSI mengilustrasikan pengiriman data untuk setiap proses dan teknologi yang dipergunakan kedalam tujuh lapis (**seven layer**).

Komponen Model OSI

(Application, Operating System & Hardware)



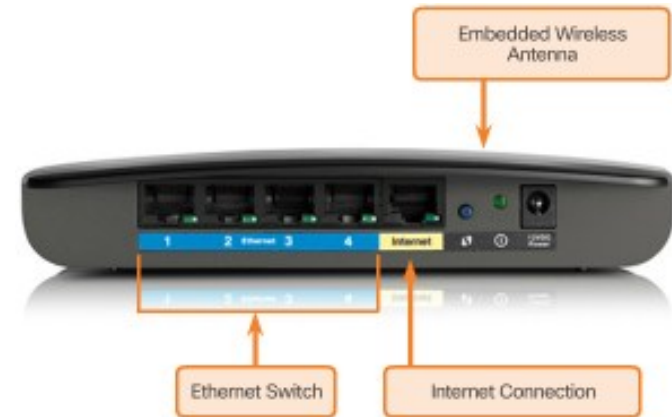
SEVEN-LAYER OSI

OSI (Open Source Interconnection) 7 Layer Model

Layer	Application/Example	Central Device/ Protocols	DOD4 Model
Application (7) Serves as the window for users and application processes to access the network services.	End User layer Program that opens what was sent or creates what is to be sent Resource sharing • Remote file access • Remote printer access • Directory services • Network management	User Applications SMTP	Process
Presentation (6) Formats the data to be presented to the Application layer. It can be viewed as the "Translator" for the network.	Syntax layer encrypt & decrypt (if needed) Character code translation • Data conversion • Data compression • Data encryption • Character Set Translation	JPEG/ASCII EBDIC/TIFF/GIF PICT	
Session (5) Allows session establishment between processes running on different stations.	Synch & send to ports (logical ports) Session establishment, maintenance and termination • Session support - perform security, name recognition, logging, etc.	Logical Ports RPC/SQL/NFS NetBIOS names	
Transport (4) Ensures that messages are delivered error-free, in sequence, and with no losses or duplications.	TCP Host to Host, Flow Control Message segmentation • Message acknowledgement • Message traffic control • Session multiplexing	PACKET FILTERING TCP/SPX/UDP Routers IP/IPX/ICMP	Host to Host
Network (3) Controls the operations of the subnet, deciding which physical path the data takes.	Packets ("letter", contains IP address) Routing • Subnet traffic control • Frame fragmentation • Logical-physical address mapping • Subnet usage accounting		Internet
Data Link (2) Provides error-free transfer of data frames from one node to another over the Physical layer.	Frames ("envelopes", contains MAC address) [NIC card — Switch — NIC card] (end to end) Establishes & terminates the logical link between nodes • Frame traffic control • Frame sequencing • Frame acknowledgment • Frame delimiting • Frame error checking • Media access control	Switch Bridge WAP PPP/SLIP	Network
Physical (1) Concerned with the transmission and reception of the unstructured raw bit stream over the physical medium.	Physical structure Cables, hubs, etc. Data Encoding • Physical medium attachment • Transmission technique - Baseband or Broadband • Physical medium transmission Bits & Volts	Hub	

Physical Layer

Tujuan dari Physical Layer untuk
Menghubungkan ke Lapisan Data Link



Connect your computer to the
Ethernet port (1, 2, 3, or 4).



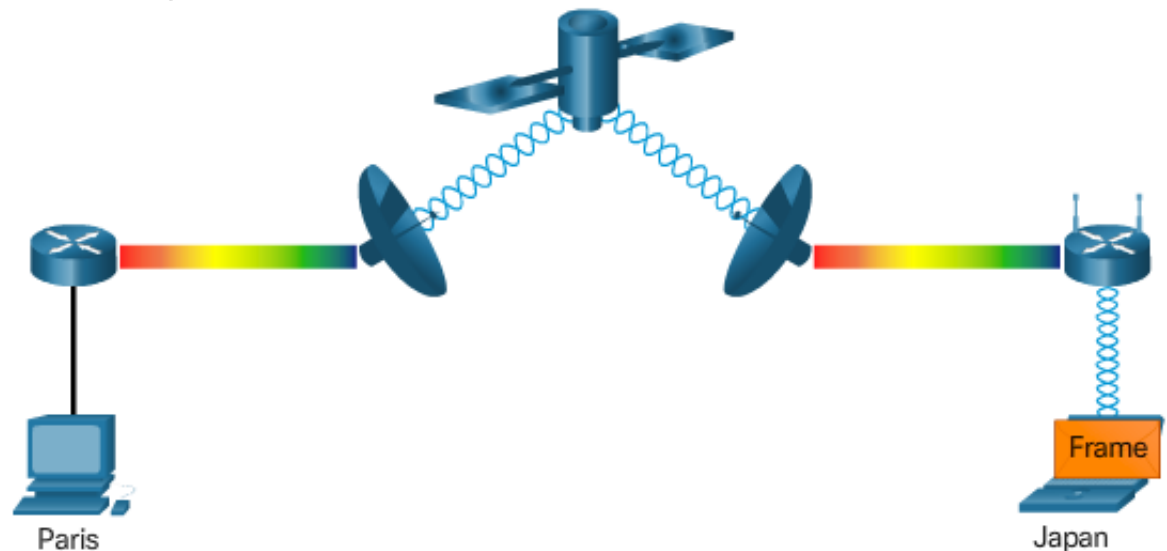
Data Link Layer

Data Link Sublayer

- LLC berkomunikasi dengan lapisan jaringan
- MAC mendefinisikan media akses proses

Standar Data Link Layer

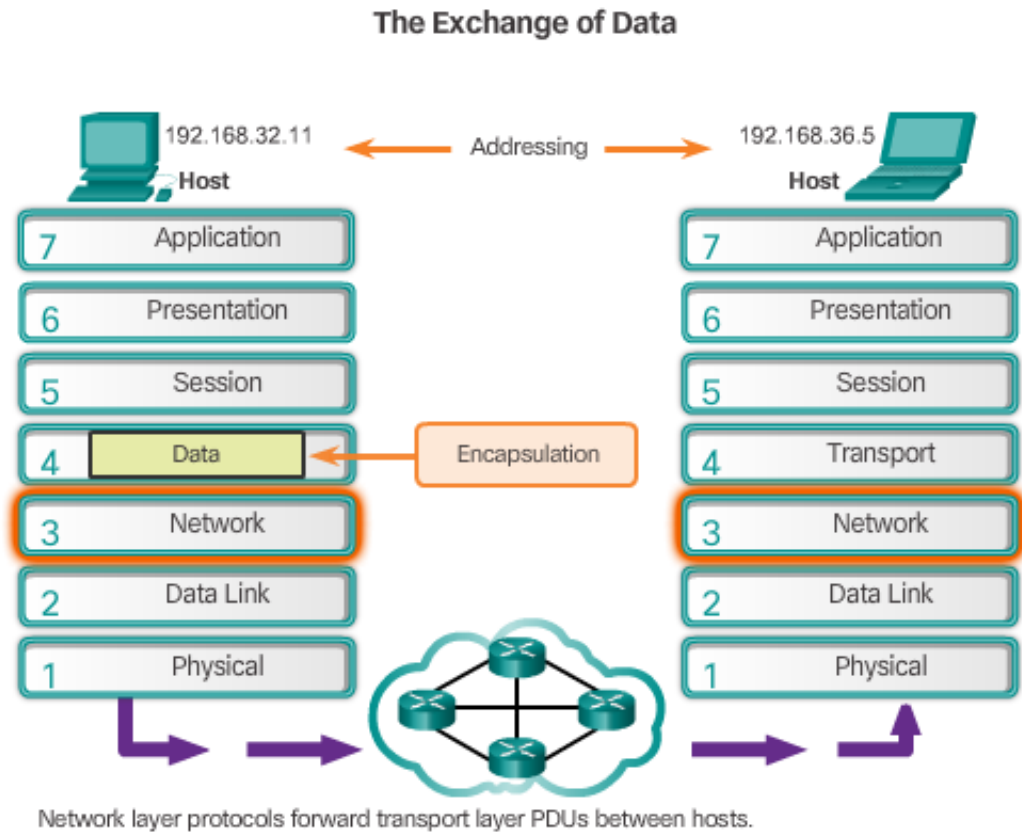
- IEEE
- ITU
- ISO
- ANSI



Network Layer

Protokol

- IPv4
- IPv6



Transport Layer

- Peran Transport Layer
 - Membangun komunikasi antara dua aplikasi dan memberikan data di antara mereka.
 - Memberikan dukungan aliran data, kontrol Flow, Multiplexing
- Kehandalan Transport Layer Protocol
 - Dua protokol yang tersedia: TCP dan UDP.
 - TCP mendukung kehandalan sementara UDP tidak.

Transport Layer

- TCP biasanya digunakan untuk,
 - Database
 - Web Browser
 - Mail, dll
- UDP
 - Live Audio
 - Video Streaming
 - VoIP, dll

Session, Presentation & Application

- **Session Layer dan Presentation Layer**
 - Session Layer menciptakan dan mempertahankan dialog antara sumber dan tujuan aplikasi.
 - Format data, kompres dan mengenkripsi data
 - Umumnya digunakan untuk video termasuk QuickTime dan Motion Picture Experts Group (MPEG).
 - Format gambar grafis umum adalah: GIF, JPEG dan PNG
- **Application Layer**
 - Lapisan yang paling dekat kepada pengguna.
 - Yang termasuk Protokol Application Layer diantaranya: HTTP, FTP, TFTP, DNS.

Application Layer Protocol (DNS)

Domain Name Service

- Dengan menggunakan DNS Server membuat alamat IP lebih mudah di hapal.
- Komputer masih memerlukan alamat numerik yang sebenarnya sebelum mereka dapat berkomunikasi.
- Protokol DNS memungkinkan untuk menerjemahkan alamat IP menjadi sebuah domain.

Application Layer Protocol (DHCP)

Dynamic Host Configuration Protocol

- Jaringan komputer memerlukan konfigurasi alamat (IP address, Subnet Mask, Default Gateway, dan lain-lain) untuk dapat berkomunikasi melalui jaringan.
- DHCP memungkinkan pemberian konfigurasi alamat pada perangkat secara otomatis.
- DHCP mendukung IPv4 dan DHCPv6 mendukung IPv6.

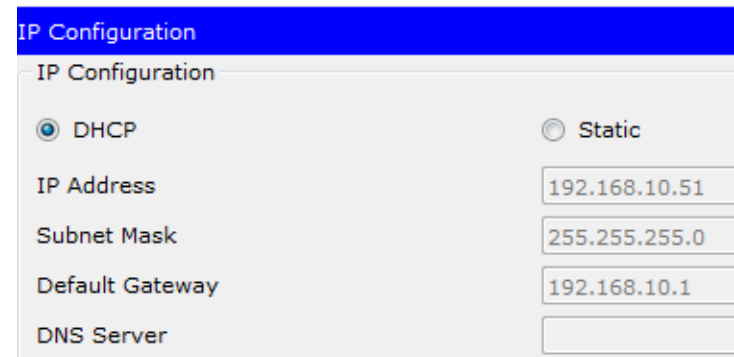
Konfigurasi Dasar DHCP Server



```
Router(config)#interface g0/0
Router(config-if)#ip address 192.168.10.1 255.255.255.0
Router(config-if)#no shut
Router(config)#ip dhcp excluded-address 192.168.10.1 192.168.10.50
Router(config)#ip dhcp excluded-address 192.168.10.100 192.168.10.254
Router(config)#ip dhcp pool JARINGAN-KOMPUTER
Router(dhcp-config)#network 192.168.10.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.10.1
Router(dhcp-config)#exit
```

DHCP Client

Untuk mendapatkan service DHCP server. Pastikan Client memilih opsi IP Configuration “DHCP” atau “automatic”



IP Configuration	
IP Configuration	
☒ DHCP	☐ Static
IP Address	192.168.10.51
Subnet Mask	255.255.255.0
Default Gateway	192.168.10.1
DNS Server	

```
C:\>ipconfig /all

FastEthernet0 Connection:(default port)

    Connection-specific DNS Suffix...:
    Physical Address.....: 0002.165C.7236
    Link-local IPv6 Address.....: FE80::202:16FF:FE5C:7236
    IP Address.....: 192.168.10.51
    Subnet Mask.....: 255.255.255.0
    Default Gateway.....: 192.168.10.1
    DNS Servers.....: 0.0.0.0
    DHCP Servers.....: 192.168.10.1
    DHCPv6 Client DUID.....: 00-01-00-01-80-14-2B-3D-00-02-16-5C-72-36
```

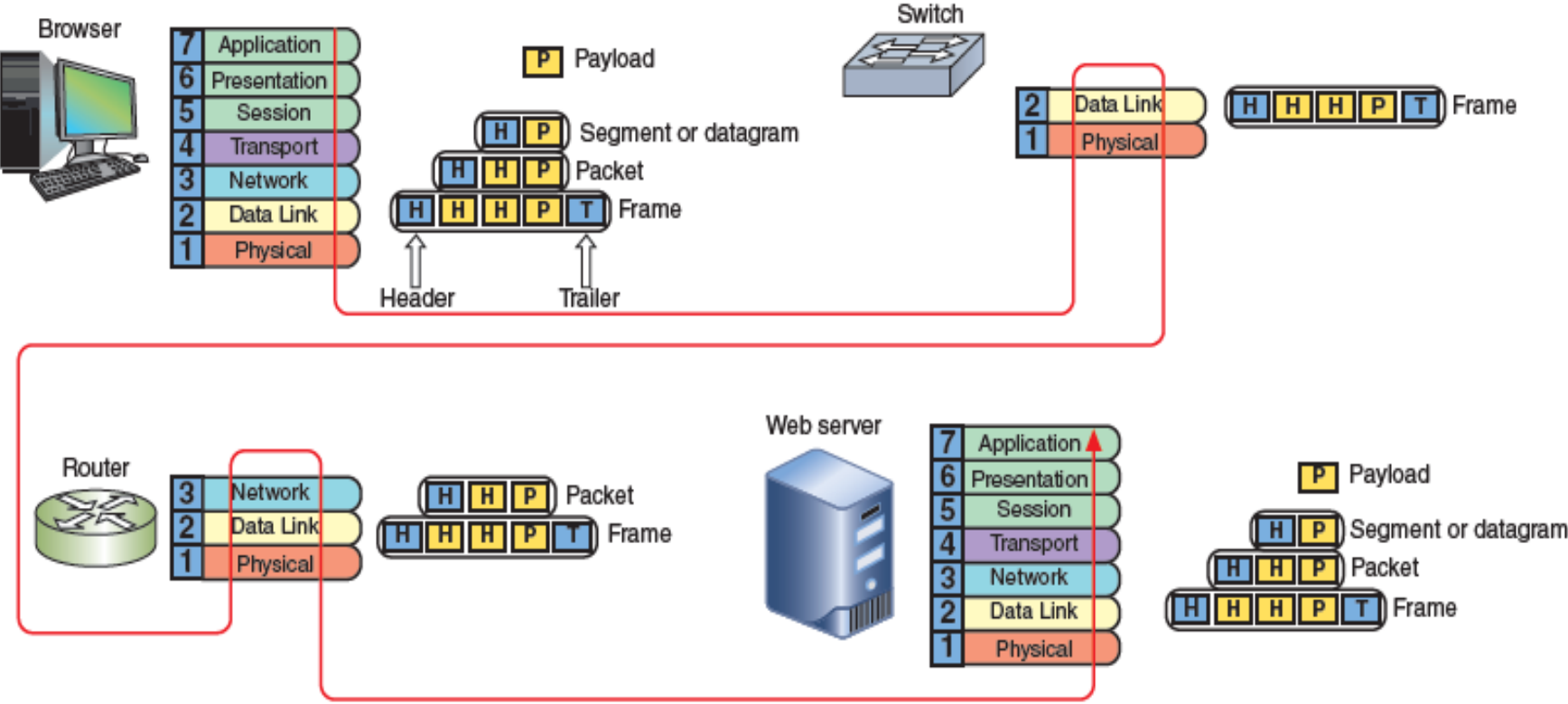
PACKET DATA UNIT (PDU)

(Penyebutan Unit Data dalam Model OSI)

OSI model	Name	Extremely technical name
Layer 7, Application layer Layer 6, Presentation layer Layer 5, Session layer	Payload or data	L7PDU
Layer 4, Transport layer	Segment (TCP) or datagram (UDP)	L4PDU
Layer 3, Network layer	Packet	L3PDU
Layer 2, Data Link layer	Frame	L2PDU
Layer 1, Physical layer	Bit or transmission	L1PDU

ALUR PROSES PENGIRIMAN DATA

(Contoh pada Layanan Web/Web Service)



Tugas Mandiri

- 1. Jelaskan Fungsi-Fungsi dari Protokol di bawah ini:
 - a) ICMP
 - b) POP3
 - c) SMTP
 - d) FTP
 - e) ARP

- 2. Berikan Penjelasan Mengenai kelebihan dan Kekurangan Ipv4 dan Ipv6?

- 3. Tugas di upload via blog masing-masing mahasiswa dan alamat blog beserta resume di email ke email dosen